

DISCUSSION OF “THE BULLWHIP: TIME-TO-BUILD AND SECTORAL FLUCTUATIONS”

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Summary

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- Provide empirical evidence for the bullwhip effect across supply chains in the US
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 - Extract downstream value-added innovations and estimate upstream IRFs
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- Quantify the importance of the bullwhip effect in accounting for supply chain volatility

MAIN FINDINGS & CONTRIBUTION

- 1 Bullwhip effect arises if current demand signals future expected demand strongly
 - hetero. lags in time-to-build + hump-shaped demand shocks amplifies upstream responses (unlike static model or uniform lags)
- 2 Even under incomplete info, hump-shaped response exists along the network, and this gets pronounced with more hump-shaped shocks
- 3 Empirically, downstream sectors exhibit AR(2) demand shocks and hump-shaped responses
- 4 The shocks also generate hump-shaped response along the supply chain, more amplified in upstream sectors, and more pronounced in sectors with more dominance of AR(2) shocks
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- ★ Tractable model giving a full analytic characterization of shock propagation
 - ★ Nicely test and quantify the time-to-build dynamics in creating bullwhip effects

Comments

OVERVIEW

- Super interesting & a very well-written paper. I learned a lot!
- Some comments:
 - ① Production Function Assumptions
 - ② Identification of Demand Shocks: Demand vs. Supply?
 - ③ Alternative sources of dynamic responses

PRODUCTION FUNCTION ASSUMPTIONS

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Q: How would allowing input substitution and endogenous time-to-build choices affect the bullwhip?

IDENTIFICATION OF DEMAND SHOCKS

Empirically, innovations in **downstream value added** (r_{it}) are treated as **demand shocks**:

$$VA_{it}^{down} = \sum_{s=1}^p \gamma_s VA_{it-s}^{down} + \mu_i + \mu_t + r_{it}$$

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 - However, value added is an equilibrium object from both supply/demand sides in the data
 - Downstream supply shocks can also propagate upstream through input demand, potentially confounding the demand-driven bullwhip effects using value-added data
- ⇒ **More direct identification or external validity (w/o model-implied estimation)** may help

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- ⇒ **More direct identification or external validity (w/o model-implied estimation)** may help
- Why demand shocks, necessarily? Can we distinguish supply vs demand shock and compare the amplification effect? (relaxing to non-unit elastic demand, e.g., CES)

ALTERNATIVE SOURCES

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- Inventory smoothing and stock-adjustment
- Convex or non-convex adjustment costs, capacity constraints
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Q: To what extent can time-to-build account for observed upstream volatility *relative to* other adjustment margins? How do these differ in welfare implications for the bullwhip?

Conclusion

CONCLUSION

This paper:

- Provides a clean, closed-form theory of the bullwhip effect + data evidence
- Highlights the role of heterogeneous time-to-build in generating shock propagation

Review: promising and well-executed paper with insightful theoretical and empirical results!

Might still be useful to

- Clarify the role of production function assumptions
- Validate the identification and distinction of demand shocks
- Discuss alternative adjustment margins and clarify how this channel differs